ProofGraph: An Exhaustive Evidence-Linked Landscape

Original query: Create an exhaustive, evidence-linked landscape for the ProofGraph concept: an AI platform that takes a question/document, extracts atomic independently verifiable claims, retrieves evidence, classifies supported/contradicted/insufficient, preserves exact provenance, models claim/source relationships and disagreement, and produces a report. Do NOT constrain research to that architecture: find all relevant materials and alternatives.

Cover: (1) research and fact-checking systems and benchmark/datasets including FEVER, FEVEROUS, AVeriTeC, SciFact, ClaimVer, FactKG, EX-FEVER, recent shared tasks, surveys; (2) retrieval: BM25/sparse, dense embeddings, hybrid fusion, reranking, multi-hop retrieval, web search, vector and lexical databases; (3) claim detection and atomic decomposition of documents and generated answers, Claimify and evaluation/faithfulness; (4) evidence verification/NLI, entailment/contradiction/neutral, sufficiency, source credibility, temporal/contextual claims, uncertainty and abstention; (5) knowledge graphs, claim/evidence graphs, provenance, contradiction links, W3C PROV, graph visualization; (6) LLM pipeline engineering, structured outputs, hallucinations, prompt injection and untrusted web content; (7) rigorous evaluation, metrics, benchmark leakage, human adjudication, calibration, baselines and ablations; (8) practical implementation: Python libraries, APIs, architecture, databases, deployment; (9) existing commercial/open-source products, adjacent tools, limitations and defensible opportunities.

For every resource distinguish category (paper, benchmark/dataset, repo, product, standard, tutorial/blog), what it provides and its limitation/relevance. Prioritize primary sources: original papers, official dataset/project pages, maintained repositories, product documentation. Include direct URLs and publication dates where available. Be clear about established techniques vs existing implementations vs genuinely novel opportunities. Avoid claiming components are novel if prior systems already do them. Surface important negative results, unresolved risks, and any recent 2025-2026 work found. Do not fabricate citations or assert completeness beyond discovered evidence.

Executive Summary

ProofGraph—an AI platform that atomically decomposes documents into verifiable claims, retrieves evidence, classifies support/contradiction/insufficiency, preserves provenance, and models claim-source relationships in a graph—sits at the intersection of five mature research streams and several nascent ones. No single prior system combines all these components identically, but each component has substantial prior art. This report maps the complete landscape: what is established, what is implemented, what is genuinely difficult, and where defensible opportunities remain as of November 2026.

Key findings for decision-makers:

1. The core pipeline (decompose retrieve classify) is well-established research. The novelty is integration: graph-structured provenance, explicit disagreement modeling, and production-grade explainability together are underserved by existing tools.

2. Retrieval quality is the dominant performance bottleneck—replacing retrieved evidence with gold evidence improves LLM accuracy by 14–22 percentage points [18].

3. Claim decomposition helps weaker verifiers but can hurt stronger ones; it is not a universally beneficial step [76].

4. Benchmark leakage (parametric and temporal) is a serious, partially unresolved threat to claimed

accuracy figures [57].

5. Prompt injection from untrusted web content is OWASP's top LLM risk (LLM01:2025) and no fully reliable defense exists [33][36].

6. Abstention and uncertainty quantification are unsolved: reasoning fine-tuning degrades abstention by ~24% on average [39].

7. The commercial market is fragmented, tool-limited, and resource-constrained—creating a real opportunity for a coherent, explainable, provenance-preserving system [83].

Section 1: Research Systems, Benchmarks, and Datasets

1.1 Core Benchmarks

FEVER (Fact Extraction and VERification) [1] remains the foundational benchmark. It contains 185,445 claims derived from Wikipedia sentences and labeled Supported, Refuted, or NotEnoughInfo (NEI), with gold evidence sentences recorded for the first two classes. Introduced at NAACL-HLT 2018 by Thorne et al. [1], it established the standard three-way classification and the pipeline of document retrieval sentence extraction verdict prediction. Its critical limitation: claims are generated by mutating Wikipedia sentences, so models can exploit claim-language bias and achieve good performance without using evidence [14].

FEVEROUS [3] extends FEVER to 87,026 claims requiring evidence from both unstructured text *and* structured table cells from Wikipedia. It introduces challenge types (Numerical Reasoning, Multi-hop, Entity Disambiguation, Combining Tables and Text), making it more realistic but harder for uniform models. Evidence is limited to three evidence sets per claim [3]. The best multi-stage retrieval pipeline (M-ReRank) achieves 93.63% Wikipedia page recall at development set but only ~10% evidence precision [81], illustrating the precision-recall gap at the retrieval stage.

AVeriTeC [2][4] is the leading real-world benchmark: 4,568 claims from 50 fact-checking organizations, labeled Supported, Refuted, Not Enough Evidence, or Conflicting Evidence/Cherry-picking, with question-answer pairs, textual justifications, speaker/publisher/date metadata, and web evidence. The 2025 shared task introduced a new scoring metric (Ev2R recall, minimum threshold 0.44 to count a verdict correct) and enforced hard reproducibility constraints: open-weights only, single GPU with 23GB RAM, 60 seconds per claim average, no proprietary LLMs [4][10]. The winning system (CTU AIC) scored 33.17% [10]; the SANCTUARY pipeline placed third at 25.27% [43]. An important design note: the AVeriTeC 2025 test set was temporally separated (train/dev claims up to 2019, test claims from 2024 onward), making it a strict out-of-distribution temporal evaluation [43].

EX-FEVER [12][13] adds multi-hop explainability: over 60,000 claims with 2-hop and 3-hop reasoning from hyperlinked Wikipedia documents, each with a veracity label and human-annotated natural language explanation of the reasoning path. Its baseline uses TF-IDF + neural retrieval (MDR), BART for explanation generation, and BERT/GEAR for verdict prediction [13]. With perfect gold explanations, BERT reaches 99.29% accuracy, suggesting explanation quality is the bottleneck [12]. Limitation: Wikipedia-only sourcing and ROUGE as the only explanation metric, which may not capture reasoning quality [12].

SciFact [17] covers scientific claim verification: 1,109 claims against a corpus of biomedical abstracts, with rationale sentences labeled Supported or Contradicted. The pipeline uses rationale selection (RoBERTa-large) followed by label prediction (fine-tuned on FEVER+SciFact). MultiVerS is the current state-of-the-art [17]. SciFact is important for ProofGraph because scientific domains require specialized

retrieval and domain-adapted NLI.

FactKG [46][51] provides 108,000 natural language claims for verification via reasoning over DBpedia knowledge graphs, covering five reasoning types: One-hop, Conjunction, Existence, Multi-hop, and Negation [51]. This is directly relevant to graph-based ProofGraph architectures. Best reported accuracy: 93.49% with single-step BERT over DBpedia subgraphs, versus 77.65% for the original GEAR-based baseline [14]. Limitation: multi-hop claims remain hardest (best 80.32% for multi-hop vs. 93.49% overall) [14]. Licensed CC BY-NC-SA 4.0, restricting commercial reuse [46].

ClaimVer [47][52] is both a system and an implicit benchmark: it verifies claims against Wikidata using multi-node BFS (Woolnet, up to 3 hops), labels as Attributable/Extrapolatory/Contradictory, and introduces a Knowledge Graph Attribution Score (KAS) [47]. Solar-10.7B-Chat achieved >89% accuracy on its custom WikiQA-derived dataset [47]. The explainability design—highlighting individual claims in a document with per-claim KG evidence—is architecturally close to ProofGraph's report generation goal.

CONFACT [29][78] addresses a gap: fact-checking with conflicting evidence. Built from 3,180 claims (566 AVeriTeC + 2,614 FactCheckQA), 17.8% exhibited conflicting evidence from sources of varying credibility [29]. Human annotators overestimated mainstream news reliability (95.8% rated credible/neutral) while MBFC flagged 30% of misleading sources that annotators trusted [29]. Inter-annotator agreement (Krippendorff's α 0.586) reflects inherent human uncertainty on conflicting evidence [29].

VERITAS [82] is the newest development (ACL 2026): a dynamic benchmark of 25,000 real-world multimodal claims from 104 organizations across 54 languages, updated quarterly to prevent leakage. It decomposes claim assessment into five dimensions (media authenticity, media contextualization, claim veracity, context coverage, integrity) each scored 1 to +1, evaluated with MSE rather than accuracy [82]. The ensemble of four frontier LLMs (GPT-5.2, Gemini 2.5 Pro, Claude Sonnet 4.5, Llama 4 Maverick) achieves MSE 0.034 against human annotations [82]. This is the most rigorous anti-leakage benchmark design currently available.

CheckThat! / CLEF [28][77] is an ongoing shared task series. The 2026 edition has three tasks: (1) source retrieval for scientific web claims, (2) fact-checking numerical and temporal claims with reasoning trace ranking, (3) generating full fact-checking articles [77]. Task 2 uses test-time scaling (multiple LLM traces at varying temperatures, de-duplicated, ranked by a verifier) [77]. The 2025 numerical claim task used BM25 over 17,000+ claims across English/Spanish/Arabic, with severe class imbalance (~58% False, 18% True, 24% Conflicting) [28].

FACTS Grounding [65] (Google DeepMind) benchmarks LLM grounding against provided input documents, with 1,719 examples covering finance, technology, retail, medicine, law, evaluated by an ensemble of three frontier LLM judges (Gemini 1.5 Pro, GPT-4o, Claude 3.5 Sonnet). The public/private split design protects against leakage [65].

FactIR [79] is a retrieval-focused benchmark from Factiverse production logs: 100 real-world claims, 90,047 corpus documents, 1,413 annotated claim-evidence pairs, zero-shot setting. BM25 and SPLADEv2 surprisingly outperform classical dense retrievers (DPR, Contriever) on this benchmark [79], with the best model (snowflake-arctic-embed-s) reaching nDCG@5 0.367 [79].

Survey resources: The TACL 2022 survey "A Survey on Automated Fact-Checking" and its EMNLP 2023 multimodal follow-up, curated in the Cartus repository [9], provide the most comprehensive cataloguing through late 2024. The 2024 survey on generative LLMs in automated fact-checking [8] covers 199 papers using PRISMA methodology and identifies multilingual and low-resource fact-checking as

underexplored areas.

Section 2: Retrieval Methods

2.1 Sparse and Lexical Retrieval

BM25 remains a strong baseline despite its age. It is used as the first retrieval stage in FEVER [1], the 2024 AVeriTeC baseline [43], ClaimCheck [7], LOKI [22], and many others. Critically, on the FactIR real-world benchmark, BM25 and SPLADEv2 *outperform* several neural dense retrievers [79]. On ClimateCheck and ClimateFEVER, TF-IDF consistently outperforms BM25, possibly because BM25's document-length normalization penalizes long passages [18]. A 2021 study found BM25 outperforms neural retrieval on the majority of 18 out-of-domain IR benchmarks [18]. TF-IDF achieves within 5% of fine-tuned transformers across 29 climate-related NLP benchmarks [18].

SPLADE-v3 (a sparse neural retriever) is used in hybrid fusion pipelines; in the ClimateCheck2025 system it is combined with BM25 and BGE-M3 via Reciprocal Rank Fusion (RRF, k=60) [24].

2.2 Dense Retrieval

Dense retrieval with FAISS inner-product indexing over sentence embeddings is used in DeReC [25], the FACT5 pipeline [11], and the journalist augmentation system [62]. Common embedding models include all-MiniLM-L6-v2 [11][58], bilingual-embedding-small, and snowflake-arctic-embed-m-v2.0 [43]. HippoRAG builds KG-based long-term memory using Personalized PageRank over dense-retrieved entity graphs [35]. DPR and Contriever underperform BM25 on domain-shifted real-world fact-checking data [79].

Multi-hop dense retrieval (MDR) is used in EX-FEVER baselines for iterative evidence accumulation across document hops [12][13]. ColBERT/SPLADE are mentioned as advanced retrievers in the ClimateCheck hybrid system [24]; ColBERTv2 is listed as a FactIR component [79].

2.3 Hybrid Retrieval and Reranking

The best-performing retrieval pipelines are consistently hybrid. The ClimateCheck2025 system fuses BM25 + fine-tuned BGE-M3 (dense) + SPLADE-v3 (sparse neural) via RRF to select top-600 candidates, then applies a cross-encoder reranker (ms-marco-MiniLM-L-6-v2) with two-phase hard negative mining, raising Recall@10 from 0.384 to 0.564 [24]. Adding an LLM-based permutation reranker (RankGPT/GPT-4.1) further raises Recall@10 to 0.601 [24].

The M-ReRank pipeline for FEVEROUS combines TF-IDF, BM25, Cross-Encoder, HybRank (sparse-dense collaborative filtering via RoBERTa), and HLATR, achieving 93.63% Wikipedia page recall [81]. SANCTUARY's two-step system uses BM25 over 500-token windows for coarse retrieval (top 125 chunks), then semantic chunking with near-duplicate filtering for fine-grained retrieval [43].

Reranking in the numerical claim task (CheckThat! 2025) did *not* improve over raw BM25 [28], illustrating domain dependency.

2.4 Web Search and Iterative Retrieval

ClaimCheck, LOKI, FIRE, and Veracity all use live web search (Serper/Google Search API) with temporal cutoff filtering to avoid post-claim evidence [7][22][44][80]. FIRE's iterative architecture uses a unified decision module: either output a verdict (when confident) or generate a new search query and

loop [44][54]. This reduces LLM cost by 7.6× and search cost by 16.5× versus fixed-N retrieval frameworks [44][54]. CONFACT's evidence was archived via the Wayback Machine to enforce temporal integrity [29].

OpenFC [40] uses a fully trained verification policy (Qwen3-8B) that decides when to retrieve, what to retrieve, and when to stop—achieving 70.39% average accuracy across six benchmarks with strict temporal cutoff enforcement [40].

Key negative result: Adding more evidence documents (3–10) to LLM context often saturated or degraded performance, attributed to the "lost in the middle" effect [28]. Noisy evidence can hurt rather than help: on ClimateCheck, adding BM25-retrieved evidence *decreases* LLM accuracy versus claim-only baselines [18].

2.5 Vector Databases

FAISS (Meta) is the most widely used [11][25][62]. ChromaDB is noted as an alternative [11][67]. Elasticsearch BM25 is used in FactIR [79]. Pinecone, Weaviate, and others are supported by the Haystack framework [70]. All are suitable for ProofGraph's evidence store component.

Section 3: Claim Detection and Atomic Decomposition

3.1 Atomic Decomposition

FActScore [5][16] is the foundational system: it breaks long-form text into atomic facts and measures the percentage supported by a reliable knowledge source. ChatGPT achieved only 58% FActScore on biographies [5]. It is available as a pip package, costing ~$1/100 sentences via OpenAI API [16]. The knowledge source is a 2023 Wikipedia dump; building the database takes ~8 hours [16]. FActScore measures *precision* only; the number of facts per response measures recall [16].

Claimify [6] (Microsoft Research, ACL 2025) is the state-of-the-art claim extraction system. It operates in four stages: sentence splitting with context creation selection disambiguation decomposition. It accepts question-answer pairs as input. Key claims: 99% of extracted claims are entailed by their source sentence; it best balances verifiable vs. unverifiable content; it is least likely to omit context critical to the fact-checking verdict [6]. Crucially, Claimify is the first system to identify when source text has multiple possible interpretations and abstain from extraction when interpretation is ambiguous [6]. **Limitation**: research-only, not commercially available [6]. Benchmarked against VeriScore, DnD, SAFE, AFaCTA, and Factcheck-GPT [6].

VeriScore [53] extracts only *verifiable* claims from long-form text, distinguishing it from FActScore which treats all claims as verifiable. It evaluates 16 models; GPT-4o is best, with Mixtral-8x22B closing the gap [53]. Factuality scores do not correlate across tasks (e.g., biography vs. long-form QA) [53], an important caution for ProofGraph system design.

VeriFastScore [49] fine-tunes Llama3.1 8B to jointly extract and verify all claims in a single pass against Google Search evidence, achieving 6.6× speedup over VeriScore. Pearson correlation with VeriScore: ~0.87 [49]. Limitation: relies on synthetic supervision from VeriScore labels; retrieval noise from sentence-level Google queries; exact-match scoring is brittle (23.7% exact vs. 71.6% with paraphrase-aware judging) [49].

SAFE [55] (Search-Augmented Factuality Evaluator, Google) decomposes responses into individual facts and verifies each via multi-step Google Search reasoning. It agrees with human annotators 72% of the

time on ~16K facts, winning 76% of 100 disagreement cases, at 20× lower cost than human annotation [55]. All code and LongFact prompts are released [55].

DecMetrics [15] introduces three decomposition quality metrics: Completeness, Correctness, and Semantic Entropy, to evaluate the output of decomposition models. Fine-tuned DeBERTa-v3-large achieves 99.14% accuracy on the synthetic test set [15]. A lightweight T5-based model trained with PPO using these metrics as a reward function is proposed [15]. **Negative result**: DecModel-large achieves ~99% correctness, but downstream fact-checking slightly underperforms non-decomposed baselines (BAcc 72.4% vs. 74.7% for MiniCheck-FT5) [15].

LOKI's pipeline [22] implements check-worthiness assessment as one of its five steps, distinguishing check-worthy from non-check-worthy claims before expensive retrieval.

ClaimBuster [64] is the pioneering check-worthiness detection tool: a machine learning model (SVM with POS, sentiment, NER features) trained on U.S. presidential debate transcripts. It provides a free API and open-source model training code, with a Docker image for self-hosting [64]. It is primarily used by journalists [64].

Document-level decomposition: LLM-CED [63] uses a single LLM across four stages (core-claim extraction, ambiguous-unit identification, QA-driven decontextualization, self-reflective review), achieving chrF 33.52% on AVeriTeC-DCE [63]. The key insight is that decontextualization—making claims self-contained—is as important as decomposition itself.

Multimodal claim extraction: MMCE [74] is a new benchmark (732 social media image-text posts) for extracting claims from multimodal content. MLLMs with image-text input outperform text-only across all metrics [74]. Entailment-based evaluation showed only moderate agreement with humans (Krippendorff's α 0.59 for reference alignment, 0.07 for entailment) [74], revealing measurement difficulty.

Decomposition Dilemmas [76]: a critical negative result. Decomposition *benefits weaker verifiers* (AlignScore) but *tends to hurt stronger ones* (Minicheck, GPT-4o-mini). Error categories: omission of context, ambiguity (vague pronouns), over-decomposition (redundancy), and meaning alteration [76]. FActScore-style decomposition produces more over-decomposition errors; VeriScore-style produces more context omission [76]. The optimal sub-claim count is at most equal to the input complexity level [76].

Section 4: Evidence Verification, NLI, and Classification

4.1 NLI Models and Classification

The standard three-way label schema (Supported/Contradicted/NEI or Entailment/Contradiction/Neutral) originates in FEVER [1] and is used across all major benchmarks. NLI-pretrained models (DeBERTa, RoBERTa) are mapped directly: entailment → Supported, contradiction → Refuted, neutral → NEI [28].

Key negative result on NLI: RefNLI [31] demonstrates that fine-tuned NLI models and few-shot prompted LLMs fail to recognize *context mismatch* between premise and hypothesis—when evidence refers to a different entity than the claim, models still predict false entailment or contradiction [31]. Human annotator agreement on this 1,143-pair benchmark reaches Fleiss' κ 0.83 [31]. Standard NLI training assumes *reference determinacy* (premise and hypothesis refer to the same context), which does not hold when evidence is retrieved from web search [31]. Filtering training examples by low lexical overlap provides only marginal improvement [31].

Reference (In-)Determinacy: Off-the-shelf RoBERTa NLI predicts far more contradictions against *irrelevant* Wikipedia pages than against the correct page [31], meaning retrieval errors compound into false contradiction predictions.

Verifier performance on benchmarks: Near-human balanced accuracy is reported on aggregated claim-verification benchmarks (LLMAggreFact, SciFact, FEVER) [66], but analysis of 24,100 GPT-4o-mini reasoning traces reveals that 49.8% of patterns are simple "direct extraction and matching," with only 0.3% genuine step-by-step verification and 10.4% multi-sentence synthesis [66]. High benchmark scores primarily reflect retrieval-plus-entailment ability, not robust verification reasoning [66].

MiniCheck (Bespoke-MiniCheck-7B) and **AlignScore-large** are commonly used as verifiers in decompose-then-verify pipelines [76]. GraphCheck [27] uses GNN-extracted knowledge graphs as soft prompts to LLMs, achieving 71.1% balanced accuracy across seven benchmarks with up to 7.1% improvement over base models [27].

4.2 Source Credibility

CONFACT [29][78] is the primary resource here: LLMs augmented with source credibility information at both retrieval and generation stages significantly improve fact-checking when evidence conflicts [78]. Annotators systematically overestimate mainstream news credibility [29], and the MBFC database (Media Bias Fact Check) is used as the credibility reference. This is an underexplored component for ProofGraph: most systems ignore source credibility entirely.

Veracity [80] shows sources with their documented credibility scores from a predefined database, providing a user-visible credibility layer. Factiverse [69] claims to rank sources by credibility using signals from reputable news media and certified fact-checkers.

4.3 Temporal Claims

ChronoFact [41][42] addresses timeline-based temporal verification: it identifies events from both claim and evidence, organizes them into chronological timelines, and predicts both the veracity of individual events and their chronological accuracy. The ChronoClaims benchmark has 40,249 training + 3,735 test claims from Wikidata [41]. 82% of ChronoFact errors involve claims with implicit temporal information [41]. Prior work (NMT, TemporalFC) was limited to structured data (tables, KGs); ChronoFact handles natural language [41].

CheckThat! 2026 Task 2 explicitly adds temporal claims to numerical claim verification [77], and OpenFC enforces strict temporal cutoffs in its web search environment [40].

4.4 Uncertainty and Abstention

AbstentionBench [39] covers 20 datasets for LLM abstention across five categories: unknown answers, underspecification, false premises, subjective questions, and outdated information. Critical findings: (1) reasoning fine-tuning degrades abstention by ~24% on average; (2) model scale has almost no effect on abstention; (3) RLVR post-training worsens abstention; (4) models often hallucinate missing context even while their reasoning chains express uncertainty [39]. A carefully crafted system prompt can boost abstention but does not resolve the fundamental inability [39].

FACT5 [11] introduces an UNVERIFIABLE abstention class as an explicit output for the ordinal fact-checking task, using it when evidence is insufficient. The pipeline reports 71.9% MSE reduction at pass@3 with Mistral-7B [11].

Calib-Abstain (preprint, low quality [38]) proposes token-level entropy + multi-path self-consistency + post-hoc calibration (temperature scaling, isotonic regression) for calibrated abstention, reporting 24–31% hallucination reduction at 80–88% coverage. Claims should be treated as preliminary given preprint status [38].

Section 5: Knowledge Graphs, Provenance, and Graph-Based Approaches

5.1 Knowledge Graphs for Fact Verification

FactKG [46][51] establishes KG-based fact verification over DBpedia, with five reasoning types. Simple one-step subgraph retrieval (all triples containing any claim entity) outperforms complex trained retrieval approaches [14].

ClaimVer [47][52] applies Wikidata BFS retrieval (Woolnet, 3-hop limit) to verify document-level claims, labeling each as Attributable/Extrapolatory/Contradictory with KG evidence citations. The KAS score aggregates triplet match scores with higher penalty for contradictory information [47]. This is the most complete prior implementation of a ProofGraph-style claim-KG verification layer.

EVOCA [32] builds AMR (Abstract Meaning Representation) sub-graphs over high-similarity claim-evidence token pairs (BERT cosine similarity threshold 0.7), converts them back to text, and passes to lightweight NLI models. Tested on Climate-BERT-fact-checking data [32]. It uses the Mehlhorn Steiner tree algorithm (via NetworkX) to extract compact sub-graphs [32].

GraphCheck [27] uses Claude-3.5-Sonnet to extract KG triples from documents and processes them with GNNs as soft prompts to LLMs, enabling fact-checking of long documents in a single inference call. It achieves up to 10.5% improvement over ACUEval, 3% over MiniCheck, and 6% over GraphEval [27]. Limitation: no reliable method exists for evaluating extracted KG quality, and gains on longer claims (AggreFact-CNN) are negative [27].

GraphRAG [50] (Microsoft) extracts knowledge graphs from raw text, builds community hierarchies via Leiden clustering, generates summaries, and uses these for RAG queries. It offers Global, Local, DRIFT, and Basic search modes [50]. Claimify was used to evaluate GraphRAG's answer comprehensiveness and diversity, showing GraphRAG outperforms standard RAG [6]. GraphRAG is an important infrastructure component for ProofGraph's knowledge layer.

CLAIRE [71] is an agentic ReAct-style system for detecting corpus-level contradictions in Wikipedia. It estimates ~3.3% of English Wikipedia facts contradict other corpus content (99% CI: 1.6%–5.0%) [71]. About 7.3% of FEVEROUS claims labeled Supported/Refuted involve corpus-level inconsistencies [71], challenging the assumption that any corpus is internally consistent. This is a critical insight for ProofGraph: even a trusted knowledge base may contain internal contradictions.

HippoRAG [35] builds a KG as long-term memory with Personalized PageRank for multi-hop retrieval, but records no provenance distinction between enrichment methods [35].

CORRECT [23] builds a three-layer evidence graph (evidence, context, reference layers) with intra- and cross-layer reasoning, and uses evidence-conditioned prompt embeddings [23]. Tested on FEVEROUS-S, BearFact, Check-COVID, and SciFact [23].

5.2 Provenance Standards

W3C PROV-DM [19] is the authoritative standard for provenance modeling. It defines entities, activities, agents, and core relations (Generation, Usage, Communication, Derivation, Attribution, Association, Delegation). It is a W3C Recommendation (stable), appropriate as the formal backbone for ProofGraph's provenance layer [19]. Key design principle: "truth of attribution ('according to X') and truth of p are logically independent" [34]—the system must distinguish sourcing from factuality.

PROV-CRED [35] extends PROV-O with enrichment method subclasses (HumanEnrichment, DeterministicEnrichment, HeuristicEnrichment, ModelEnrichment), a provcred:Model class (identity, version, task type, benchmark performance), and a ReliabilityAttribution [35]. Open problem: confidence scores from KGE models, LLMs, and heuristics are not on a common scale; no standard cross-method calibration dataset exists [35]. LLMs that score their own triples exhibit correlated errors (self-evaluation bias) [35].

RDF-Star (forthcoming in RDF 1.2) provides syntactically clean triple-level annotation without reification, but defines no semantics for the annotations [35]. RDF 1.2 will make ProofGraph's provenance layer more implementable at triple granularity.

Provenance-Enhanced Statements [34] introduces the DEC framework, which interprets provenance predicates as epistemic stance indicators and groups provenance-homogeneous statements into cognitive worlds using doxastic/epistemic modal logics [34]. The prototype reasoner is implemented on Apache Jena/Fuseki [34]. This is academically rigorous but may be over-specified for a first production system.

Wikidata's approach [34]: reified statements with provenance, qualifiers, and ranks (preferred/normal/deprecated), where a statement can be promoted to factual by asserting it again as an unreified triple. This is a practical reference implementation.

5.3 Claim-Evidence Graph Modeling and Contradiction Links

The CONFACT pipeline models source credibility as a node attribute in the retrieval graph [29][78]. EVOCA aligns claim and evidence AMR graphs [32]. FactKG uses KG paths as evidence structures [46]. The hybrid pipeline in [20] models a one-hop KG retrieval layer with web fallback, using entity linking (ReFinED) and cross-encoder scoring (MiniLM) for triple relevance [20].

No prior system has explicitly modeled **contradiction links as first-class graph edges** between claims and sources, or between competing sources—this is a genuinely underserved design element of ProofGraph.

Section 6: LLM Pipeline Engineering

6.1 Structured Outputs

Structured JSON output is "well-supported mainly in GPT-4" [22]; switching to Llama-3 can cause pipeline failures without specific design [22]. LOKI wraps all LLM calls with a retry mechanism and traffic queue [22].

Constrained decoding [72] eliminates structural failures (raises schema validity to 100%) in small LLMs (0.6B–4B) under Outlines and XGrammar, but a persistent semantic gap remains—schema conformance does not guarantee content accuracy [72]. XGrammar adds only 4–8ms compilation overhead vs. 2–5s for Outlines [72]. Constrained decoding can push generation toward locally valid but semantically incorrect trajectories [72].

DSPy [11] provides declarative prompt optimization that replaces manual prompt engineering, using chain-of-thought at each step—used in the FACT5 pipeline [11].

6.2 Hallucination Mitigation

Hallucinations arise from stale training data, probabilistic language modeling, and knowledge conflicts between retrieved evidence and model parameters [59][56]. Key finding: when retrieved evidence conflicts with LLM internal knowledge, the LLM may ignore the retrieved evidence and produce hallucinations [56].

DeReC [25] argues that explicit rationale generation may not be needed for high verification accuracy—dense retrieval + DeBERTa classifier achieves competitive F1 with 92–95% less runtime vs. LLM rationale pipelines [25]. This trades interpretability for efficiency.

Chain-of-thought prompting with explicit citation of Q-A pairs before label production (SANCTUARY [43]) improves groundedness. GraphCheck's GNN attention weights provide evidence-level explainability [27] .

Knowledge conflicts: LLMs with internal knowledge of a claim may "help" verify it without evidence (zero-evidence GPT-4o achieves 53.2% on AVeriTeC) [7], meaning parametric leakage contaminates even retrieved-evidence systems.

6.3 Prompt Injection and Untrusted Web Content

This is the most serious unresolved risk for ProofGraph when ingesting web evidence.

Threat severity: OWASP rates prompt injection as LLM01:2025, the top LLM vulnerability [33][36]. Indirect prompt injection occurs when untrusted external content (web pages, files) alters LLM behavior without user awareness [33]. Five carefully crafted documents can manipulate AI responses 90% of the time via RAG poisoning [33].

State of defenses (2025–2026): CaMeL (Google DeepMind, March 2025) uses Privileged + Quarantined LLMs with data provenance tracking, solving 77% of AgentDojo tasks vs. 84% undefended [36]. FIDES (Microsoft Research, May 2025) applies information-flow control, stopping all attacks in internal testing [36]. Meta's LlamaFirewall reduced attack success rate by ~90% to 1.75% [36]. MELON (ICML 2025) achieved 0.32% ASR on AgentDojo at double the API calls [36].

However: 12 published defenses were bypassed at >90% success rate by adaptive attacks [36]. The International AI Safety Report 2026 found sophisticated attackers bypass best-defended models ~50% of the time with 10 attempts [36]. Anthropic states even adversarially trained Claude Opus 4.5 at ~1% ASR "still represents meaningful risk" [36]. **Prompt injection cannot be fully solved within current LLM architectures** per OpenAI, Anthropic, and Google DeepMind 2025 publications [36].

Practical defenses for ProofGraph: The Dual LLM pattern—a Privileged LLM for trusted pipeline instructions and a Quarantined LLM without tool access for untrusted web content [37]; Microsoft Spotlighting (input delimiting, datamarking, encoding, reportedly reducing ASR from >50% to <2%) [37]; Llama Guard for input/output classification [37]; architecture design to minimize blast radius of successful injection [37]. ClaimCheck's ethics section explicitly acknowledges the risk of inadvertently retrieving and summarizing low-quality or misleading sources [7].

Note: Neither ClaimCheck [7], VeriFastScore [49], LOKI [22], nor most research systems implement described defenses against prompt injection from web content—this is an open gap in the field.

Section 7: Rigorous Evaluation, Metrics, and Leakage

7.1 Metrics

Standard classification metrics: Macro-F1 is most widely used [56]. Accuracy can diverge from macro-F1 by up to 20 points under class imbalance [18]. FEVER score combines verification accuracy with evidence retrieval accuracy [56]. The FEVEROUS score requires both correct label *and* adequate evidence coverage [81].

Evidence retrieval metrics: Recall@k, nDCG@k, B-Pref (incomplete-judgment-aware) [24]. Ev2R recall (atomic reference-based evidence coverage, threshold 0.44 for AVeriTeC 2025 [4]) is the most rigorous current standard, replacing the brittle Hungarian METEOR metric [43].

Factuality metrics: FActScore (atomic precision) [5], VeriScore (verifiable claim precision) [53], SAFE F1 [55], VeriFastScore (correlated 0.87 Pearson with VeriScore) [49]. FACTS Grounding uses LLM ensemble judging (3 frontier models) [65]. MSE is proposed for ordinal/continuous scoring [11][82].

Decomposition quality: DecMetrics (Completeness, Correctness, Semantic Entropy) [15]; ClaimScore (supported atomic claims) criticized for missing semantic overlap [15].

Abstention: AbstentionBench uses abstention recall as primary metric (models generally have high precision, so recall is the bottleneck) [39].

Inter-annotator agreement: RefNLI achieves Fleiss' κ 0.83 [31]; CONFACT Krippendorff's α 0.586 [29]; MMCE entailment α 0.07 (very low) [74]; FactIR Cohen's κ 0.946 [79]. The low agreement on entailment-based evaluation of extracted claims is a warning for automated evaluation design.

7.2 Benchmark Leakage

This is a critical risk for any evaluation of ProofGraph or its components.

Parametric leakage: LLMs encode training-corpus knowledge of claims. Masking named entities on AVeriTeC collapses supported-claim recall from 0.421 to 0.131 while overall accuracy drops only 0.03 points—aggregate accuracy hides parametric leakage [57]. Zero-evidence GPT-4o achieves 53.2% accuracy on AVeriTeC [7]. GPT-4o web search "benefits from temporal data leakage" and cannot be constrained by publication cutoff dates, frequently retrieving annotated fact-check articles [7].

Temporal leakage: Allowing post-claim evidence raises zero-shot accuracy by 6.3 points on AVeriTeC [57]. 18.8% of AVeriTeC evidence answers with trusted on-page dates post-date the claim [57]. Live search-engine date filters still return post-cutoff pages [57]. The AVeriTeC 2025 test set fixes a temporal leakage issue in the 2024 document collection [4].

Data contamination: Evaluation data can enter pre-training corpora and inflate reported scores [11][57]. VERITAS addresses this with quarterly updates and a held-out private set [82].

Annotation leakage in MultiFC: Evidence often comes from the same fact-checking article that states the ruling, allowing models to copy rather than verify [57].

Mitigation strategies: Temporal cutoff filtering in retrieval [7][62][40]; holding out private test sets [65]; fresh data collection [11]; the new Ev2R metric requiring evidence coverage not just label correctness [4][43]; VERITAS's quarterly update commitment [82].

7.3 Calibration

NN-PPI [67] calibrates LLM check-worthiness predictions via nearest-neighbor residual correction from a small labeled calibration set, without retraining. It improves weighted F1 by up to 33.80% for small models (Gemma 3 4B) on ClaimBuster [67]. Larger commercial models show near-zero gain (already saturated) [67]. The system uses all-MiniLM-L6-v2 embeddings indexed in ChromaDB [67]. Standard post-hoc calibrators (Platt scaling, temperature scaling, isotonic regression) are most effective when miscalibration is homogeneous; NN-PPI targets input-dependent bias [67].

Confidence scores from different model families are not on a common scale; no standard cross-method calibration dataset exists [35].

7.4 Human Adjudication

Professional journalists evaluated an automated fact-checking tool and reported only 59% of claims accurately classified and 58% of retrieved evidence sentences relevant [83]. Inter-rater reliability between major fact-checking organizations on factual accuracy is very low [62]. Free-text explanations improve lay users' detection of false information but not journalists' [83]. These findings suggest that ProofGraph's evaluation must include professional user studies, not just automated metrics.

Section 8: Practical Implementation

8.1 Key Python Libraries and APIs

Component	Tool/Library	Notes
Retrieval (sparse)	Elasticsearch BM25, BM25Okapi (rankbm25)	Low latency, easy deployment [79]
Retrieval (dense)	FAISS [25][11], ChromaDB [67], Pinecone, Weaviate [70]	FAISS for flat/HNSW; ChromaDB for smaller scales
Retrieval (sparse neural)	SPLADEv2 [79], BGE-M3 [24]	Strong on domain-shifted data
Reranking	ms-marco-MiniLM cross-encoders [20][24], RankGPT [24]	Cross-encoder: expensive but accurate
Embedding models	all-MiniLM-L6-v2 [11][58][67], snowflake-arctic-embed [43][79]	MiniLM: fast; arctic-embed: strongest FactIR
NLI/verifier	DeBERTa-v3-large [15][28], MiniCheck [76], AlignScore [76]	DeBERTa for classification; Bespoke-MiniCheck-7B for grounding
Claim decomposition	FActScore pip package [5][16], Claimify (research only) [6], VeriScore [53]	FActScore: easiest to install; Claimify: best quality
Check-worthiness	ClaimBuster API [64], NN-PPI (open) [67]	ClaimBuster: free API; NN-PPI: post-hoc calibration layer
Web search	Serper API [7][22][44], Google Programmable Search [20], Bing API [62]	Serper: $0.001/search; SerpAPI: $0.015/search [22]
Graph/KG	NetworkX [32], Apache Jena/Fuseki [34], GraphRAG [50]	Jena for SPARQL/provenance; GraphRAG for community summaries
Pipeline orchestration	Haystack [70], DSPy [11], LangChain	Haystack: Kubernetes-ready, hybrid retrieval, audit controls
LLM access	OpenAI API, Fireworks AI (serverless), vLLM (local) [63][43]	vLLM: quantized local deployment; Fireworks: cost-effective

Structured output	Outlines, XGrammar [72]	XGrammar: negligible overhead; Outlines: larger compilation cost
Prompt injection defense	NeMo-Guardrails, Llama Guard, LLM Guard [37]	Defense-in-depth required; none is fully reliable
Provenance standard	W3C PROV-O [19], PROV-CRED extensions [35]	RDF-Star (RDF 1.2) for triple-level annotation

8.2 Reference Architecture for ProofGraph

Based on the evidence, a production ProofGraph system would minimally include:

1. **Input layer**: Document/question ingestion → sentence segmentation → decontextualization (LLM-CED-style [63] or Claimify-style [6])
2. **Claim detection**: Check-worthiness scoring (ClaimBuster [64] or NN-PPI-calibrated SLM [67]) → filtering non-verifiable claims (VeriScore-style [53])
3. **Retrieval layer**: Hybrid BM25 + dense (FAISS/snowflake-arctic-embed) + sparse neural (SPLADE) → Reciprocal Rank Fusion [24] → cross-encoder reranking → temporal cutoff enforcement [7][40]
4. **Evidence quarantine**: Quarantined LLM for untrusted web content processing [37]; Spotlighting/delimiting for injection resistance [37]
5. **Verification layer**: NLI/verifier (DeBERTa or Bespoke-MiniCheck-7B) for each claim-evidence pair; confidence scoring; abstention when below threshold [11]
6. **Provenance graph**: PROV-O-based graph [19] with PROV-CRED enrichment → method annotation [35]; claim nodes, evidence nodes, source nodes, verdict nodes; contradiction links as first-class edges
7. **Source credibility**: MBFC-style credibility database lookup per source [29]; credibility as edge/node attribute in provenance graph
8. **Report generation**: Verdict aggregation per claim (Supported/Contradicted/Insufficient); structured JSON with provenance pointers; human-readable explanation generation (Llama-3.1-8B-Instruct style [58])
9. **Deployment**: Haystack pipelines (serializable, Kubernetes-ready) [70]; FastAPI backend; PostgreSQL for structured provenance; FAISS/ChromaDB for vector evidence store; Next.js frontend [80]

Cost estimate (inference): FIRE-style iterative retrieval costs ~$0.01–0.02 per atomic claim [48]; VeriFastScore estimates $1/100 sentences [49]; LOKI estimates $0.001/search via Serper [22].

Section 9: Commercial and Open-Source Products

9.1 Open-Source Systems

System	Type	Key Features	Limitations
LOKI/OpenFactVerification [22][45]	Open-source tool/repo	5-step pipeline, multilingual (EN/ZH), async, JSON output, Serper retrieval, MIT license, 1.2k GitHub stars	GPT-4 required for reliable structured output; research performance, not production-hardened
FIRE [44][54]	Research framework/repo	Iterative retrieval, 7.6× LLM cost reduction, agent-style, open LLM support, NAACL 2025	27 GitHub stars; small community; FacTool-QA/FELM/BingCheck benchmarks only

Veracity [80]	Open-source app	Next.js/FastAPI/PostgreSQL/GCP/Kubernetes, credibility scores, expert dashboard, user feedback	Proprietary LLM still required; no prompt injection defenses described
FActScore [16]	Python package	pip install, $1/100 sentences, Wikipedia knowledge base, abstain detection	Wikipedia-only knowledge source; 2023 cutoff; no provenance graph
SciFact pipeline [17]	Research code	RoBERTa rationale selector + label predictor, COVID-19 report generation, AWS S3 models	Scientific domain only; 2020-era models
EX-FEVER [13]	Dataset + baseline code	Multi-hop explainability, BART explanations	Low adoption (12 GitHub stars); Wikipedia-only
ClaimBuster [64]	API + open ML code	Free API, Docker image, check-worthiness, presidential debate training data	Political domain bias; older SVM-based approach
DecMetrics [15]	Research code	Decomposition quality metrics, T5-based decomposer with RL	Synthetic training data; research-stage
GraphRAG [50]	Open-source library	KG extraction, Leiden clustering, community summaries, multiple query modes	Requires prompt tuning; config changes between versions
Haystack [70]	Orchestration framework	Hybrid retrieval, self-correction loops, Kubernetes-ready, multi-provider, audit controls	Framework layer only; no built-in fact-checking logic

9.2 Commercial Products

Product	Type	Key Features	Limitations
Factiverse [69]	Commercial SaaS	40+ language claim detection, Google/Bing/Wikipedia live search, credibility ranking, API, €10/month	3–10 word optimal claim length; best on news topics only; no KG/provenance
Full Fact AI [68]	Nonprofit + commercial	Monitors millions of sentences daily, 40 countries EN/FR/AR, used in TPFC	Meta ended TPFC program Jan 2025; under funding pressure; limited public API
Factiverse (GPT plugin)	Plugin	ChatGPT integration with credible source citations	Plugin ecosystem limitations; reliance on ChatGPT

9.3 Adjacent Research Systems

OpenFC [40] (Qwen3-8B post-trained verification policy) achieves best overall averages (70.39% accuracy, 63.30% macro-F1) across six benchmarks, treating the entire verification workflow as a learned policy—the most integrated research system as of 2026, but not yet packaged for deployment.

ClaimCheck [7] is the highest-accuracy system on AVeriTeC dev (76.4%, Qwen3-4B), using real-time Google Search with temporal cutoffs, but is a demo-stage system.

SANCTUARY [43] (3rd place AVeriTeC 2025) is a complete, reproducible pipeline with detailed design documentation and vLLM deployment—one of the best references for a production-quality fact-checking system architecture.

ClaimVer [47][52] (EMNLP 2024 Findings) is the closest prior implementation of a claim-KG verification

layer with per-claim explainability and attribution scores.

Section 10: Negative Results, Open Problems, and Opportunities

10.1 Important Negative Results

1. **Retrieval degrades accuracy under domain shift**: Adding BM25-retrieved evidence *decreases* LLM accuracy on ClimateCheck vs. claim-only [18]. Noisy evidence is worse than no evidence for strong verifiers.

2. **Decomposition can hurt stronger verifiers**: Decompose-then-verify improves weaker models but hurts GPT-4o-mini and MiniCheck [76].

3. **Abstention worsens with reasoning fine-tuning**: RLVR/RLHF reasoning training degrades abstention ~24% on average; scaling does not help [39].

4. **NLI context mismatch**: Models produce false entailment/contradiction when evidence refers to a different entity [31].

5. **LLM rationale generation adds hallucination risk**: LLM-generated explanations frequently contain factual inconsistencies and lack grounding [25].

6. **Dense retrieval underperforms BM25 on domain-shifted real-world fact-checking** [79][18].

7. **Adding more evidence hurts via "lost in the middle"** [28].

8. **No reliable prompt injection defense exists** for agentic web-retrieval systems [36].

9. **Corpus internal consistency cannot be assumed**: ~3.3% of Wikipedia facts contradict other corpus content; 7.3% of FEVEROUS claims involve corpus-level inconsistencies [71].

10. **Inter-annotator agreement on entailment evaluation of extracted claims is very low** (α 0.07 [74]), making automated evaluation unreliable for complex claim types.

10.2 Open Problems

• **Multilingual and low-resource fact-checking** are systematically underexplored [8][56]; most tools are English-only or English-centric.

• **Multimodal claim extraction** is largely unexplored [74]; over one-third of debunked claims involve images and text [74].

• **Conflicting evidence and source credibility integration** remain immature [29][78]; most systems assume a single best evidence source.

• **Cross-benchmark generalization**: the best system on SciFact (macro-F1 0.700) drops to 0.315 on ClimateCheck [18]; systems do not generalize reliably across domains.

• **Calibration across heterogeneous sources**: confidence scores from NLI models, LLMs, and KG retrievers are not on a common scale [35].

• **Multi-hop KG retrieval**: multi-hop evidence retrieval from KGs is still a major challenge [20]; best models score 80.32% on multi-hop vs. 93.49% overall on FactKG [14].

• **Implicit temporal information**: 82% of ChronoFact errors involve claims with implicit temporal cues [41].

• **Explanation quality evaluation**: ROUGE inadequately measures reasoning quality in explanations [12]; entailment-based evaluation has low human agreement [74].

• **Retroactive provenance for large KGs**: bootstrapping enrichment method records for

Wikidata/DBpedia from edit logs is an open problem [35].

• **Dynamic leakage prevention**: even quarterly-updated benchmarks like VERITAS [82] face the challenge that the most recent quarter is "generally the only one suitable for reliable benchmarking."

• **Journalist adoption**: fact-checkers are skeptical of AI tools and report only 59% claim accuracy and 58% evidence relevance in prior evaluations [83].

10.3 Genuinely Novel Opportunities for ProofGraph

The following elements are individually established in research but *not integrated* in any existing open-source or commercial system:

1. **First-class contradiction link modeling**: explicitly representing claim-source disagreements as typed graph edges (not just labeling each claim independently). No prior system does this [as analyzed from 29][78][71].

2. **Provenance-aware source credibility as a graph property**: combining PROV-O/PROV-CRED enrichment method annotation [35] with real-time source credibility scoring [29][78] in a queryable provenance graph—ProofGraph could make the credibility of each evidence link explicit and auditable.

3. **Confidence-calibrated abstention at claim level**: combining NN-PPI calibration [67] with UNVERIFIABLE abstention [11] and AbstentionBench insights [39] into a per-claim uncertainty output that feeds into the provenance graph. No existing system provides calibrated uncertainty *per claim* with provenance.

4. **Cross-corpus consistency checking as a verification step**: using CLAIRE-style [71] detection of internal contradictions within the evidence corpus to flag when evidence for a claim is inconsistent with other known evidence—treating corpus inconsistency as a first-class verification signal.

5. **Temporal claim timeline modeling integrated with provenance**: applying ChronoFact's [41] timeline structure within a provenance graph, linking each claimed event to temporally-indexed evidence and tracking when contradictions arise in timelines.

6. **Journalist-facing explainability with claim-level audit trail**: the user study literature [83][62] is clear that professional fact-checkers need per-step transparency, source selection rationale, and bias criteria. No current tool provides this at the level ProofGraph envisions. The business case is real: fact-checking organizations are under severe funding pressure [83][68] and need tools that make professional workflows faster without replacing editorial judgment.

7. **Defense-in-depth prompt injection architecture**: implementing the Dual LLM pattern [37] (Privileged LLM for trusted pipeline instructions, Quarantined LLM for web content) combined with Spotlighting [37] and taint tracking—no current research fact-checking system describes this in its architecture, representing a genuine security-first design opportunity.

Summary Table: ProofGraph Component Coverage

ProofGraph Component	Established Research	Best Prior Implementation	Gaps/Opportunities
Atomic claim decomposition	FActScore, Claimify, VeriScore, SAFE	Claimify (ACL 2025)	Ambiguity flagging at scale; multimodal claims
Check-worthiness	ClaimBuster, NN-PPI	ClaimBuster API	Calibration in deployment; domain shift
Hybrid retrieval	BM25+dense+SPLADE+RRF	ClimateCheck2025 pipeline	Real-world fact-checking still brittle;

		[24]	domain shift
Iterative retrieval	FIRE, OpenFC, ClaimCheck	OpenFC (best overall accuracy)	Injection-resistant iterative agents
NLI/verdict classification	DeBERTa, MiniCheck, GraphCheck	Bespoke-MiniCheck-7B	Context mismatch; sufficiency; calibration
Source credibility	CONFACT, Factiverse	CONFACT (research)	No production system integrates this
Temporal claims	ChronoFact	ChronoFact (IJCAI 2025)	Implicit temporal cues; provenance linking
Abstention/NEI	AbstentionBench, FACT5-UNVERIFIABLE	FACT5 pipeline	Reasoning models degrade abstention
Knowledge graph evidence	FactKG, ClaimVer, GraphRAG	ClaimVer (EMNLP 2024)	Multi-hop; production-scale
Provenance modeling	W3C PROV-DM, PROV-CRED	PROV-CRED (research)	No production system uses PROV-CRED fully
Contradiction links	Conceptually present in CONFACT	None as first-class graph objects	Genuine design opportunity
Corpus consistency	CLAIRE	CLAIRE (arxiv 2025)	Integration with verification pipeline
Structured output	Constrained decoding (XGrammar)	XGrammar [72]	Semantic gap persists in small models
Prompt injection defense	CaMeL, FIDES, LlamaFirewall	None in research fact-checking	Major unresolved risk
Report generation	JustiLM, QRAFT, CheckThat! Task 3	QRAFT (GPT-4o multi-agent)	LLM articles still lag human writers
Calibration	NN-PPI, temperature scaling	NN-PPI [67]	No common scale across model types
Benchmark leakage control	VERITAS quarterly, Ev2R	VERITAS [82]	Most benchmarks remain leakage-vulnerable

Watchlist for Decision-Makers

Monitor closely (2026-2027):

- AVeriTeC 2026 shared task results (tracking reproducible, efficient open-weight systems)
- CLEF CheckThat! 2026 Task 3 (fact-checking article generation quality trend)
- VERITAS quarterly updates (first dynamic multimodal benchmark; track leakage-resistant evaluation)
- RDF 1.2 finalization (RDF-Star enabling clean triple-level provenance annotation)
- CaMeL/FIDES/MELON prompt injection defense maturity (currently best academic solutions but not production-ready)
- OpenFC/Qwen3-based learned verification policies (may supersede hand-engineered pipelines)

Key risks:

- Prompt injection from web content is an existential risk for any web-retrieval fact-checking system;

it cannot be fully mitigated today [36]

• Parametric and temporal leakage inflate accuracy figures across the board; treat all published numbers with appropriate skepticism [57]

• LLM abstention degrades with the same fine-tuning that improves reasoning; this directly affects ProofGraph's NEI classification reliability [39]

• Corpus inconsistency means even trusted knowledge bases cannot be treated as ground truth [71]

Defensible unique value: ProofGraph's combination of (1) provenance-annotated claim-evidence graph with contradiction links, (2) source credibility as a first-class graph attribute, (3) calibrated per-claim uncertainty, (4) journalist-facing audit trail, and (5) defense-in-depth prompt injection architecture—if executed together—represents a genuinely differentiated position in a market where existing tools are fragmented, unexplainable, and resource-constrained [83][68].

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